

A 4- λ $\times$ 32-Gb/s Silicon Micro-Ring-Resonator-Based DWDM Receiver with On-Chip Temperature Controller

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Abstract We demonstrate a 4- λ $\times$ 32-Gb/s silicon MRR-based DWDM receiver with an on-chip temperature controller, automatically aligning and locking each MRR resonance wavelength to its target wavelength. Its operation is confirmed by electronic-photonic co-simulation using the Cadence design environment and experimentally verified under thermal stress. ©2026 The Author(s)

Introduction

As demands for interconnect systems with high bandwidth density, low power consumption and low latency for AI computing in data centers continue to grow, optical I/O has become an essential solution. In particular, the micro-ring resonator (MRR) has shown great potential to achieve optical I/O scale up due to its compact size and dense wavelength division multiplexing (DWDM) compatibility [1]. But its characteristics are highly sensitive to fabrication process and temperature variations, requiring wavelength calibration methods to align and lock each MRR resonance wavelength to the desired input wavelength [2-3].

Various techniques have been reported to implement temperature controllers (TCs) within an electronic IC (EIC) [4-5], which provide accurate wavelength calibration for MRRs. To optimize the operation of TCs, the electronic-photonic co-simulation is necessary during the design stage.

However, electronic-photonic co-simulation has been limited by simulator capabilities and modeling methodologies.

In this paper, we demonstrate the 4- λ $\times$ 32-Gb/s silicon MRR-based DWDM receiver with the on-chip TC by integrating our previously reported FPGA-based TC into the EIC, which automatically aligns and locks all WDM channels through a single scanning process [6]. To evaluate the operation of the TC together with the photonic integrated circuit (PIC), co-simulations using Cadence photonic device models are performed with optical frequency analysis (OFA) and transient simulations in the Cadence Spectre simulator. Measurement results demonstrate that the fabricated DWDM receiver with the on-chip TC successfully operates at the data rate of 32-Gb/s against external thermal stress by automatically adjusting the optimal heater voltage (V_H) to each MRR.

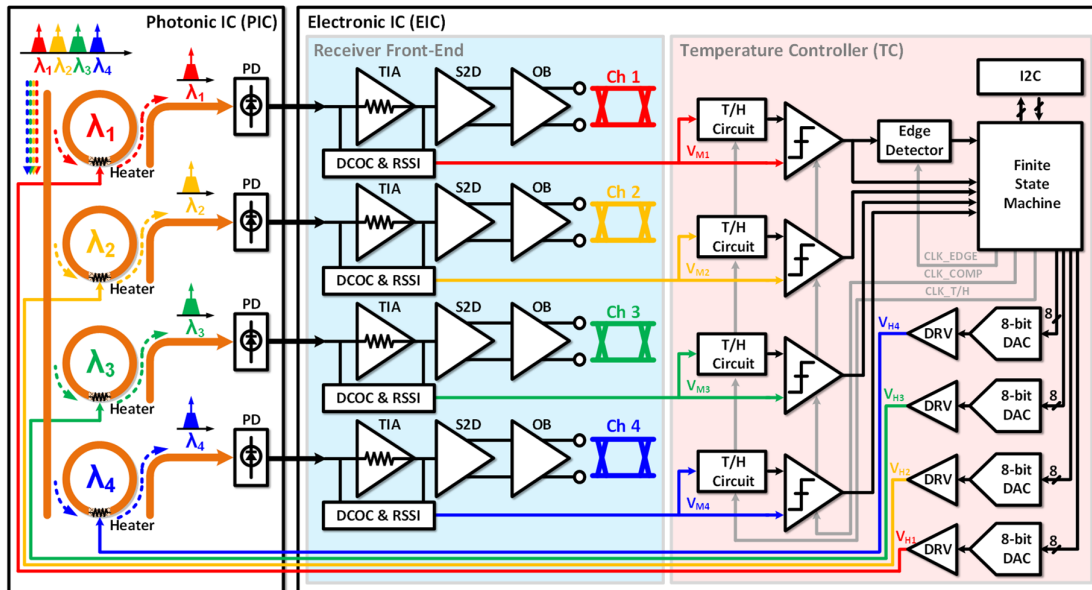


Fig. 1: Block diagram of the DWDM receiver with the on-chip temperature controller.

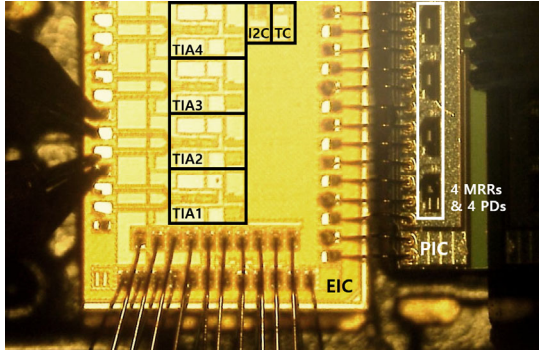


Fig. 2: Chip photograph of the fabricated DWDM receiver.

System Implementation

Fig. 1 shows the block diagram of the DWDM receiver with the on-chip TC and its chip photograph is shown in Fig. 2.

The PIC is fabricated by AMF and contains four cascaded MRRs having the same radius of 10- μ m. Each MRR has a metal heater for thermal tuning of its resonance wavelength and a germanium photodetector (PD) at its drop port, whose anode is wire-bonded to the EIC.

The EIC is fabricated in 28-nm complementary metal-oxide-semiconductor (CMOS) technology by Samsung Foundry. In the EIC, the receiver front-end contains four transimpedance amplifiers (TIAs), four single-to-differential amplifiers (S2Ds) and four output buffers (OBs). Each TIA is equipped with a DC offset cancellation (DCOC) loop and a received signal strength indicator (RSSI) that monitors received average input optical power. The TC contains four track-and-hold (T/H) circuits, four comparators, an edge detector, a finite state machine, four 8-bit digital-to-analog converters (DACs) and four metal heater drivers (DRVs). The TC operates in two modes: a scan mode and a dither-and-track mode. Initially, in the scan mode, the TC sweeps the heater voltage (V_H) of one MRR by increasing the 8-bit DAC code. DAC codes at which the RSSI monitoring voltage (V_M) reaches its peaks are detected by the edge detector, stored, and then applied to each MRR. In this way, initial wavelength calibration time can be reduced. During the dither-and-track mode, T/H circuits and comparators compare the previous and current V_M values. Based on the comparison result, V_H is determined to maximize V_M for each WDM channel, corresponding to the maximum optical power delivered to the drop port.

Electronic-Photonic Co-Simulation and Measurement Results

To evaluate the performance of DWDM receiver with the on-chip TC in the design stage, electronic-photonic co-simulations of the TC together with the PIC are performed. For the PIC behavior

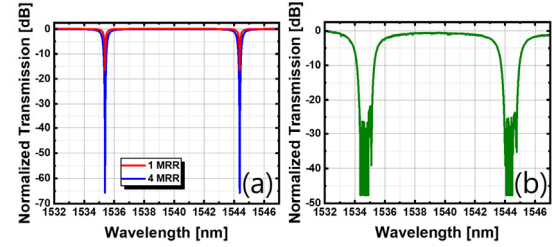


Fig. 3: (a) Simulated and (b) measured transmission characteristics for MRRs at the bus waveguide output.

emulation, Cadence photonic device models are used. With these models, the transistor-level TC operation can be simulated within the Cadence environment.

For measurements, the fabricated EIC and the PIC are packaged on an FR4 PCB and mounted on a temperature-controllable chip stage. Four different input wavelengths of $\lambda_1 = 1536.5$ -nm, $\lambda_2 = 1537.3$ -nm, $\lambda_3 = 1538.1$ -nm and $\lambda_4 = 1538.9$ -nm are combined using a 4 by 1 coupler and only one input wavelength can be modulated using a Mach-Zehnder modulator (MZM) with 32-Gb/s pseudo-random binary sequence (PRBS) data due to the limitation of measurement setup. An erbium-doped fiber amplifier (EDFA) is used to compensate for the insertion loss of the MZM, followed by a variable optical attenuator. Combined four different input wavelengths are injected into the PIC through a fiber array. The receiver front-end circuits are supplied with a 1.2-V supply voltage and heater drivers are supplied with a 2-V supply voltage, which enables resonance wavelength thermal tuning beyond one free spectral range of each MRR.

Fig. 3 shows simulated and measured transmission characteristics of the cascaded MRRs at the bus waveguide output. For the simulation, each Cadence photonic MRR model (rr_double-bus) is configured with the same radius of 10- μ m used in the fabricated PIC. Compared with the simulated transmission, the measured transmission shows broader resonance wavelength distributions caused by process variation.

Fig. 4 shows simulated and measured V_M and V_H values during the scan mode of the TC operation. In both the simulation and measurement, the MRR close to the input optical coupler in PIC is used to scan four different input wavelengths. The TC increases V_{H1} and measures V_{M1} to detect required V_H values that make V_{M1} peaks. Then, the desired V_H value to each MRR is assigned and the dither-and-track mode starts.

To evaluate the TC operation, thermal stress is applied to the fabricated DWDM receiver by changing the chip stage temperature sinusoidally from 17- $^{\circ}$ C to 33- $^{\circ}$ C with a period of 720 seconds as shown in Fig. 5(a). Fig. 5(b) shows measured V_H values in response to thermal stress.

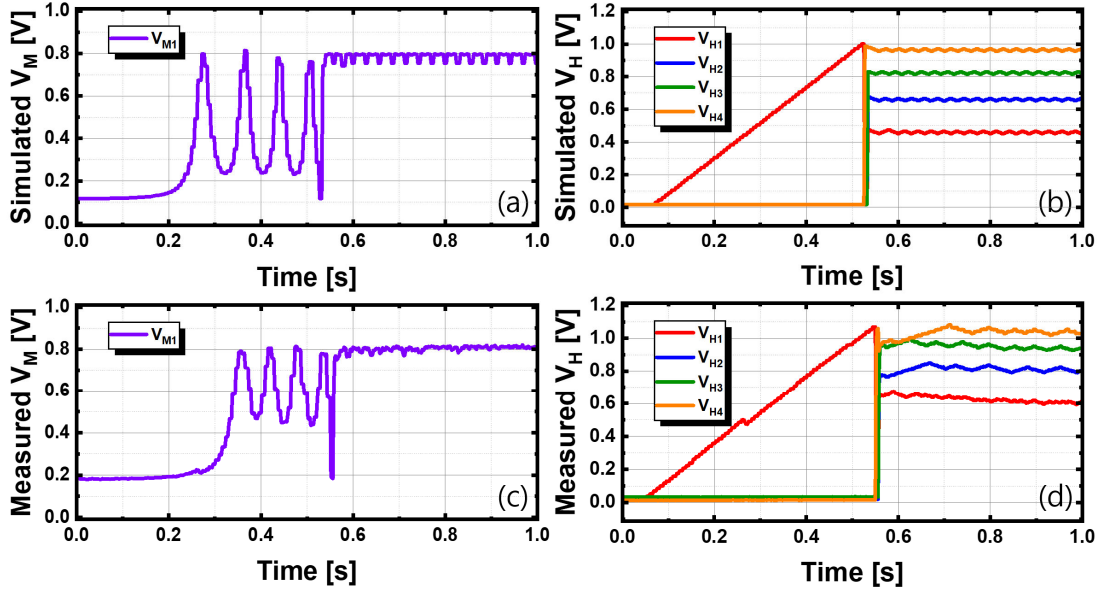


Fig. 4: (a) Simulated V_M , (b) V_H , (c) Measured V_M and (d) V_H values during the scan mode of the temperature controller.

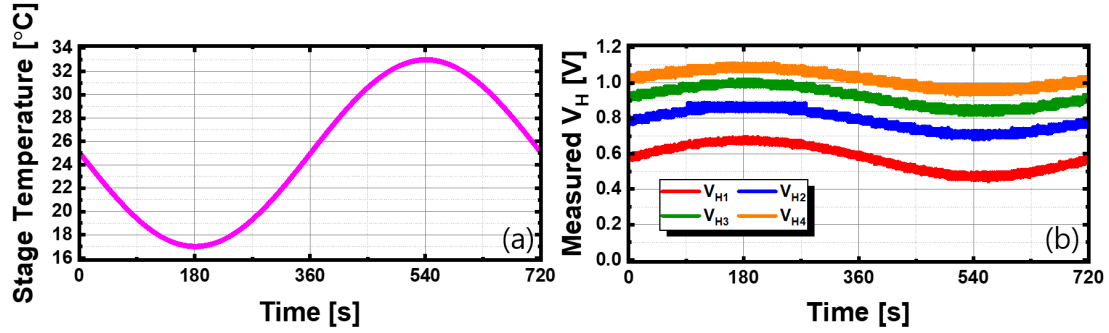


Fig. 5: (a) Measured stage temperature and (b) V_H values during the dither-and-track mode under thermal stress.

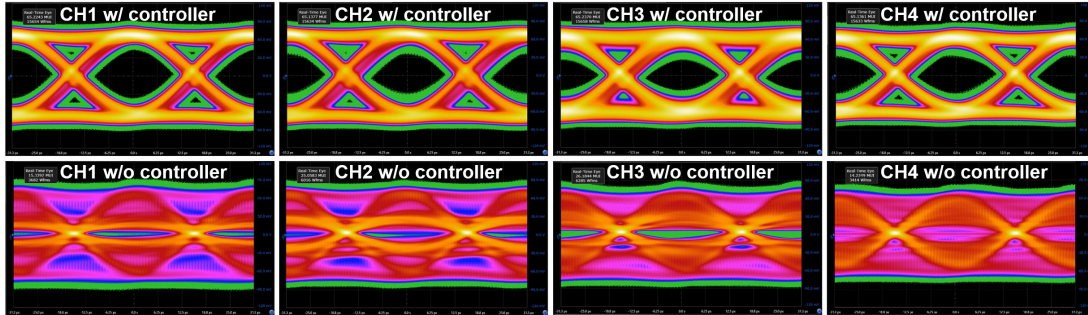


Fig. 6: Measured eye diagrams of each DWDM channel under thermal stress, with and without the temperature controller.

Fig. 6 shows measured accumulated eye diagrams of each DWDM channel under thermal stress with and without the TC. For these measurements, one WDM channel is modulated with 32-Gb/s PRBS-23 data at -1-dBm average optical power, while the other DWDM channels are set to -1-dBm DC optical power. As can be seen in the figure, the TC maintains the good-quality eye diagrams by automatically adjusting the V_H of each MRR under thermal stress.

Conclusions

We demonstrate the 4- λ x 32-Gb/s silicon MRR-based DWDM receiver with the on-chip TC which automatically searches for and maintains the optimal heater voltage of each MRR against thermal stress. In addition, the TC design is carried out with electronic-photonic co-simulation using Cadence photonic device models within the Cadence environment.

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